

Similarity Index of Cellular Automata

Erick Jesse Angeles Lopez

Mentor: Sukanta Das

August 28, 2026

Cellular automata have been used to model complex systems [1, 2, 3], from biology to economics [4, 5]. For a single problem there may be several cellular automata capable of simulating it and, the more complex the system, the more alternatives appear when trying to simplify it. Something similar happens with the Game of Life [6]: The rule Life proposed by John Conway coexists with a family of rules, called *life-like*, that exhibit a similar behavior. But how similar are they? Is there a rule that approximates it more, and another that approximates it less? The goal of this research is to define a similarity measure that assigns a numerical value to how similar two given rules are based on their evolution, instead of relying solely on whether they “look” alike visually.

As a first result, we identified four relations that connect rules from different definitions, that is, cellular automata that differ not only in the rule number but also in the neighborhood or the number of states: state permutation and mirror rules (well-known relations that reduce the analysis space; in ECA, from 256 rules to only 88 [7]), together with two new relations that map similarity across definitions: neighbor reduction (when the value of a neighbor does not have any effect in the transition function, it can be removed and the behavior is the same but with a smaller neighborhood) and state reduction (when a state is never produced and it only appear in the initial condition). By applying these four reductions iteratively, it is possible to build a connection graph in which the shortest path between two rules measures how far apart they are; this shows that there exist rules whose behavior is the same even in different spaces.

The next methods build the transition graph of the cellular automaton, where each node represents a space configuration and each edge the transition from that configuration to the next one. By obtaining the canonical form, it is possible to decide whether two rules are structurally identical while ignoring node

labels, but it only offers a binary comparison, so it does not work as a similarity measure.

The third method computes the eigenvalues of the adjacency matrix of the graph and uses them as a feature vector, measuring the cosine distance between two rules. However, it only works for rules within the same space: the number of eigenvalues depends on the number of nodes, so it is not possible to compare, for example, rules with 2 and with 3 states.

To fix this problem we defined the spectral moments method: given an n value, the system is evolved n steps and at each step we count the proportion of configurations that returned to their original state. This count vector has a fixed length that depends on the chosen range of steps and not on the number of nodes of the graph, so it does allow comparing rules from different spaces using the cosine distance.

We applied this spectral moments method to Conway’s Game of Life in two experiments. On an exact toroidal 4×4 grid (16 cells) we compared the standard Life rule (B3/S23) with two Moore-neighborhood variants, Life+S8 (B3/S238) and Life+B8 (B38/S23); Life and Life+B8 produced identical vectors (cosine distance 0), while Life+S8 differed slightly from both (distance 0.0005), which suggests that modifying the survival condition has a greater structural impact than modifying the birth condition. In a second experiment, on a toroidal 6×6 grid (36 cells, where the exact graph is infeasible due to its 2^{36} configurations) we sampled 3000 initial configurations per rule, with a 200-step transient and a 150-step limit to detect the return period, comparing Life, HighLife (B36/S23), Seeds (B2/S) and two Von Neumann neighborhood rules (B2/S12 and B13/S2). Four of the five rules, the three Moore rules and Von Neumann B2/S12, grouped together with cosine distances between 0.0001 and 0.0065, while Von Neumann B13/S2 was clearly isolated from the rest (distances between 0.056 and 0.095): the detected similarity does not depend only on the neighborhood type, but on the particular dy-

namics of each rule. Additionally, we compared this experiment on the raw configuration space against the same method grouping configurations by rotations: for 4 of the 5 rules both produce nearly identical vectors (distance below 0.00023), which suggests that quotienting by rotations is a safe computational simplification that does not compromise the discriminative power. The exception is again Von Neumann B13/S2, whose raw-versus-quotiented distance was larger (0.026) and which also showed a notable fraction (23.7%) of samples whose return period was not resolved within the considered step limit.

We also applied the spectral moments to the 256 ECA rules (rotation-quotiented graph, 14 cells, 10 steps): only 63 vectors are distinct. Clustering shows 14 groups with Euclidean distance and 3 with cosine distance (one of them concentrates the well-known chaotic and complex rules), but the distinction is not sharp (silhouette 0.254; 71 rules have a neighbor from another group closer than 0.01): it is more a continuous spectrum than well-separated classes.

Finally, we tested the measure on a scenario taken from a different line of research: adaptive evolution in totalistic cellular automata [8], which studies the rules that produce finite lifeforms (patterns that, starting from a single active cell, grow and then die out completely) and propose that all of them are connected through single-digit mutations of the rule. In that space ($k = 3$, $r = 1$) we confirmed the connectivity, the 45 rules (out of 729) with a finite lifeform form a single connected component, but their spectral cosine distance shows no trend with the mutation distance (Spearman 0.006), even pairs with identical evolution reach distances of 0.276. The reason: lifeforms analyze a single initial condition, while the spectral moments consider all of them.

It remains to determine the minimum space size, or context window, that preserves the distinctive information between rules without doing unnecessary computational costs, a question that arises from noticing that, for elementary cellular automata, a context of only 4 cells is already enough to distinguish almost all known behaviors. It is also necessary to extend the validation of the measure to other dimensions and space sizes, for example grids larger than 4×4 or 3-dimensional cellular automata, to confirm that it generalizes beyond the 2D case studied. In particular, more 2D experiments are required to contrast what was observed in the first Life experiment: the suggestion that modifying the survival conditions has a greater structural impact than modifying the birth conditions comes from a single exact 4×4 grid, evidence far too small for such a general claim, so it

must be verified on larger grids using the sampling technique already employed (random initial configurations with a step transient and a limit to detect the return period) over more birth and survival variants. The cases whose return period is not resolved within the considered step limit remain to be investigated, which could indicate chaotic or very-long-period dynamics. Finally, since the transition graph grows exponentially with the number of cells and states, optimization methods are required, like more efficient sampling, spectral approximations, or further graph reductions, so that this similarity measure can be applied to large-scale cellular automata.

References

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